DATA COMMUNICATIONS & NETWORKS — QUIZ 1

Module: Lecture 1 Part 1 (Fundamentals, Switching & Evolution)

Name: _____ Student ID: _____ Score: _____ / 20

1. Which of the following is NOT an appropriate definition or criterion of data communication?

- ◯ a. Sharing digital information according to agreed communication protocols.
- ◯ b. Exchanging data between two or more connected devices.
- ◯ c. Transmitting information through a physical or unguided communication medium.
- ◯ d. Delivering data accurately and reliably to an intended destination.
- ◯ e. Processing data entirely within one isolated computer system without external transfer.

2. Which statement best describes the fundamental physical nature of wireless communication?

- ◯ a. Data is transmitted without utilizing any electromagnetic waves or physical signals.
- ◯ b. Data communication does not require encoding, modulation, or antennae.
- ◯ c. Data can only be transmitted through bound copper cables or optical fibers.
- ◯ d. Data is carried by electromagnetic waves propagated through space without a physical wired medium.
- ◯ e. Wireless communication is exclusively limited to light in the visible spectrum.

3. Which technology is specifically designed for short-range personal-area communication rather than long-distance telecommunications?

- ◯ a. Satellite communication
- ◯ b. 5G Cellular network
- ◯ c. Bluetooth
- ◯ d. Public Switched Telephone Network (PSTN)
- ◯ e. Television broadcasting

4. In an early telephone network, a full-mesh topology is deployed for 10 subscribers. Calculate the total number of physical direct lines needed so every subscriber connects directly to every other subscriber.

Hint: Number of links $= \frac{n(n-1)}{2}$

Write your final numerical answer:

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5. A regional branch office network expands from $n_1 = 6$ workstations to $n_2 = 12$ workstations. Assuming a full-mesh topology is maintained at all times, how many *additional* communication links must be installed?

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6. Which switching methodology serves as the fundamental technical backbone of the modern global Internet?

- ◯ a. Circuit switching
- ◯ b. Packet switching
- ◯ c. Message switching
- ◯ d. Dedicated crossbar switching
- ◯ e. Frequency-division fixed line switching

7. Traditional analog landline telephone systems (PSTN) historically relied upon which type of switching?

- ◯ a. Packet switching
- ◯ b. Store-and-forward switching
- ◯ c. Circuit switching
- ◯ d. Datagram switching
- ◯ e. Cell relay switching

8. Which statement is FALSE regarding the historical Advanced Research Projects Agency Network (ARPANET)?

- ◯ a. ARPANET was funded by the United States Department of Defense (DoD).
- ◯ b. ARPANET was initially designed and operated as a wide-area circuit-switched telephone network.
- ◯ c. ARPANET pioneered packet switching concepts across geographically distributed nodes.
- ◯ d. ARPANET was instrumental in developing and adopting the TCP/IP protocol suite.
- ◯ e. ARPANET initially interconnected computing centers at universities and research laboratories.

9. Which statement best articulates the formal difference between *telecommunication* and *data communication*?

- ◯ a. Telecommunication is analog-only, while data communication involves only optical fiber.
- ◯ b. Telecommunication strictly excludes voice, whereas data communication includes all voice streams.
- ◯ c. Telecommunication broadly covers communicating over long distances, whereas data communication specifically targets digital information exchange between computing devices.
- ◯ d. Telecommunication operates without protocols, whereas data communication uses protocols.
- ◯ e. Telecommunication requires wired infrastructure, whereas data communication is strictly wireless.

10. What major breakthrough in the early 1990s propelled the Internet from an academic/military utility into a globally accessible medium for everyday consumers?

- ◯ a. The commercial deployment of transatlantic copper telegraph cables.
- ◯ b. The invention of the World Wide Web (WWW) and graphical web browsers.
- ◯ c. The initial introduction of rotational telephone dials.
- ◯ d. The standardization of Ethernet 10BASE5 thicknet.
- ◯ e. The migration to 2G GSM cellular networks.

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11. Which characteristic best exemplifies the operational paradigm of the modern "Mobile and Cloud Era"?

- ◯ a. Software applications and databases run exclusively on local, offline workstations.
- ◯ b. Processing and storage are centralized in remote data centers, accessible ubiquitously across varied mobile devices.
- ◯ c. Network architectures have abandoned standard Internet Protocols in favor of proprietary links.
- ◯ d. Devices must remain statically tethered to local physical wiring to receive updates.
- ◯ e. Computing workloads are restricted strictly to mainframe batch-processing jobs.

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12. Why is a full-mesh topology impractical for large-scale public communication networks involving hundreds of thousands of users?

- ◯ a. Mesh topologies do not allow data encryption.
- ◯ b. The number of physical transmission links scales quadratically ($O(n^2)$), causing astronomical cabling costs and port exhaustion.
- ◯ c. Mesh networks create a single point of failure where one severed link halts the entire network.
- ◯ d. Mesh networks can only transmit analog audio signals and cannot carry digital packets.
- ◯ e. Data packets in a mesh topology can only travel in one unidirectional direction.

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13. In circuit-switched networks, what occurs during the communication process before any user data can be transferred?

- ◯ a. Packets are partitioned, tagged with sequence numbers, and sent independently across random paths.
- ◯ b. A dedicated end-to-end physical or logical circuit must be established and reserved.
- ◯ c. Intermediate switches store the entire message on disk before forwarding.
- ◯ d. All other active telephone calls across the switch are immediately terminated.
- ◯ e. The system dynamically negotiates bandwidth on a per-packet basis.

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14. What primary efficiency benefit does packet switching offer over circuit switching for typical bursty computer data traffic?

- ◯ a. It guarantees zero transmission delay and zero jitter.
- ◯ b. It dynamically shares transmission capacity among multiple sessions via statistical multiplexing when users are idle.
- ◯ c. It avoids needing any network protocols or destination addresses.
- ◯ d. It eliminates the need for buffers and memory at intermediate routers.
- ◯ e. It permanently reserves channel bandwidth regardless of whether data is transmitting.

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15. If a tier-1 Internet Service Provider (ISP) interconnects 15 autonomous routing centers in a fully meshed core topology, how many physical core links are required?

Show your formula setup:

$$\text{Links} = \frac{n(n-1)}{2}$$

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16. The electrical telegraph, invented in the 19th century, represents an early milestone in data communication primarily because:

- ◯ a. It relied on optical lasers transmitting digital packets across continents.
- ◯ b. It encoded alphanumeric characters into discrete binary-like pulse patterns (dots and dashes).
- ◯ c. It functioned as the world's first wireless packet-switching mesh network.
- ◯ d. It was the first system to implement TCP/IP layer abstraction.
- ◯ e. It carried human analog voice signals without distortion.

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17. Which of the following is considered a *guided* (wired) communication medium?

- ◯ a. Terrestrial microwave
- ◯ b. Unshielded Twisted Pair (UTP) copper cable
- ◯ c. Satellite uplink/downlink
- ◯ d. Infrared line-of-sight
- ◯ e. Cellular radio waves

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18. What was the principal reason ARPANET developers selected packet switching over circuit switching?

- ◯ a. Circuit switching was too modern and unproven at the time.
- ◯ b. To build a robust, fault-tolerant network where traffic could dynamically reroute around damaged nodes or lines.
- ◯ c. Because packet switching was cheaper for analog telephone calls.
- ◯ d. Packet switching eliminated the need for computers to have network interfaces.
- ◯ e. Packet switching allowed televisions to receive digital broadcasts.

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19. How has the transition from traditional telecommunications to modern converged IP networks impacted telephone calls?

- ◯ a. Telephony has moved from packet switching to dedicated analog circuits.
- ◯ b. Voice calls are digitized and packetized (e.g., VoIP), traveling over the same packet-switched data networks as web and email traffic.
- ◯ c. Telephone calls now bypass communication protocols entirely.
- ◯ d. Voice communication has been replaced entirely by telegraph codes.
- ◯ e. Voice calls now require full-mesh physical cables between callers.

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20. What critical standard protocol adoption occurred on January 1, 1983 (known as "Flag Day"), marking the birth of the modern Internet architecture on ARPANET?

- ◯ a. Replacement of NCP (Network Control Program) with TCP/IP.
- ◯ b. Introduction of HTTP/1.0 and HTML.
- ◯ c. Replacement of fiber optics with coaxial cable.
- ◯ d. Conversion of the Internet into a circuit-switched telephone grid.
- ◯ e. Adoption of 802.11 Wi-Fi standards.

ANSWER KEY & DETAILED EXPLANATIONS

Lecture 1 Part 1 — Solutions

Question 1: e Data Communication Concept

Explanation: Data communication fundamentally involves transmitting encoded data between two or more remote/connected devices over a communication medium. Processing data inside an isolated standalone computer is local data processing, not data communication.

Question 2: d Transmission Media

Explanation: Wireless communication transmits data by modulating electromagnetic waves (such as radio frequencies or microwaves) propagating through the atmosphere or vacuum, requiring no bound physical conductor.

Question 3: c Network Scope

Explanation: Bluetooth is an IEEE 802.15 standard engineered specifically for short-range Personal Area Networks (PANs, typically ≤ 10 meters). Cellular, satellite, telephone, and TV systems are long-distance telecommunication systems.

Question 4: 45 links Mesh Topology Calculation

Explanation: In a full mesh topology with $n = 10$ nodes:

$$\text{Links} = \frac{n(n-1)}{2} = \frac{10(10-1)}{2} = \frac{10 \times 9}{2} = \frac{90}{2} = 45 \text{ links}$$

Question 5: 51 links Scaling Calculation

Explanation: For $n_1 = 6$:

$$\text{Links}_1 = \frac{6 \times 5}{2} = 15 \text{ links}$$

For $n_2 = 12$:

$$\text{Links}_2 = \frac{12 \times 11}{2} = \frac{132}{2} = 66 \text{ links}$$

$$\text{Additional Links} = 66 - 15 = 51 \text{ links}$$

Question 6: b Switching Types

Explanation: The Internet is built entirely upon **packet switching**, where data streams are broken into discrete datagrams/packets routed independently across shared links.

Question 7: c PSTN Infrastructure

Explanation: Traditional telephone networks (PSTN) utilized **circuit switching**, which reserves a physical or dedicated logical circuit between two endpoints for the entire duration of the call.

Question 8: b ARPANET History

Explanation: ARPANET was expressly designed as the world's first operational wide-area **packet-switched** network, explicitly rejecting circuit switching due to vulnerability to single-point line failures.

Question 9: c Terminology

Explanation: Telecommunication ("tele" = at a distance) covers any transmission of voice, video, or signals across distances. Data communication is a specialized subset focusing on digital data exchange between computational nodes.

Question 10: b Internet Milestones

Explanation: The development of the World Wide Web (invented by Tim Berners-Lee) along with user-friendly graphical browsers (such as NCSA Mosaic and Netscape) made the Internet understandable and accessible to non-technical everyday users.

Question 11: b Computing Eras

Explanation: In the mobile and cloud computing era, services, applications, and storage live in ubiquitous cloud infrastructure and are accessed dynamically across diverse endpoints (smartphones, tablets, laptops) from any location.

Question 12: b Topology Complexity

Explanation: A full mesh requires $\frac{n(n-1)}{2}$ links. For 100,000 users, that would require $\approx 5 \times 10^9$ dedicated lines, which is physically and financially impossible. Networks therefore use hierarchical switching.

Question 13: b Circuit Switching

Explanation: Circuit switching requires three distinct phases: Circuit Establishment (connection setup), Data Transfer, and Circuit Teardown. No user data flows until the dedicated path is confirmed.

Question 14: b Statistical Multiplexing

Explanation: Computer communication is typically bursty (idle time interspersed with rapid bursts). Circuit switching wastes reserved capacity during idle periods; packet switching dynamically shares line capacity among all active users.

Question 15: 105 links Core Network Math

Explanation:

$$\text{Links} = \frac{n(n-1)}{2} = \frac{15(15-1)}{2} = \frac{15 \times 14}{2} = \frac{210}{2} = 105 \text{ links}$$

Question 16: b Historical Milestones

Explanation: The telegraph was the earliest practical digital communication system because it translated human language into discrete binary-like symbols (short dots and long dashes) transmitted electrically.

Question 17: b Guided Media

Explanation: Guided media uses physical conductors or guides (twisted pair copper wire, coaxial cable, fiber optic strands). Terrestrial microwave, satellites, infrared, and cellular use unguided (wireless) air/space.

Question 18: b ARPANET Robustness

Explanation: A core requirement from the US Department of Defense (DoD) was survivability: if some nodes or links were severed during an attack, packet-switched routers could dynamically reroute surviving packets over alternate paths.

Question 19: b Convergence

Explanation: Voice over IP (VoIP) samples and digitizes analog audio into standard IP packets, allowing telephony to converge onto shared, standardized packet-switched data networks.

Question 20: a Internet Architecture

Explanation: On January 1, 1983, ARPANET officially switched its underlying core communication protocol from the original NCP to TCP/IP, creating the foundational internetworking protocol structure used to this day.